Regression Discontinuity: The Scholarship Example

Experimentation, A/B Testing and Causal Inference (SDA-DSC-213), SDAIA Academy. Handout for Day 3, Hour 4 (slides 113 to 115). All five Day 3 handouts use the same story: a tutoring programme at one school.

Step 1. The story

Students who score **70 or more** on the entrance test receive a **scholarship**. Those below 70 do not. We want the effect of the scholarship on first-year GPA. Scholarship holders have higher GPAs, but they were also better students to begin with.

Symbol	Meaning	Here
R	running variable	entrance test score
c	cutoff	70
T	treatment	scholarship, $T = 1$ if $R \geq 70$
Y	outcome	first-year GPA

Step 2. The idea (slide 113)

A student who scored 69 and one who scored 70 are practically the same student; which side of 70 they landed on is close to luck. Comparing students just below and just above the cutoff is a local experiment. The jump in GPA at 70 is the estimate estimand: the effect at the cutoff.

Step 3. The data, in score bins

Entrance score	Scholarship	Students	Mean GPA
66	no	20	2.85
67	no	22	2.90
68	no	21	2.95
69	**no**	**20**	**3.00**
70	**yes**	**21**	**3.35**
71	yes	20	3.40
72	yes	22	3.45
73	yes	19	3.50

Step 4. The estimate

Simplest version: a narrow bandwidth of one point on each side.

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τ̂ = mean GPA at 70 − mean GPA at 69 = 3.35 − 3.00 = +0.35
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Better version: fit a straight line on each side within a bandwidth and take the difference of the two lines **at** the cutoff. GPA rises 0.05 per score point on both sides, so the left line, continued to 70, predicts 3.05, and the right line at 70 is 3.35.

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left line at 70: 3.00 + 0.05 = 3.05 right line at 70: 3.35 τ̂ = 3.35 − 3.05 = +0.30
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The estimate estimate: +0.30 GPA points. The simple 69-versus-70 comparison gave 0.35 because it also contains one point of the underlying slope; fitting lines removes that.

Report the estimate across several bandwidths (slide 113): a wide bandwidth uses more students but risks bias from the curve; a narrow one is unbiased but noisy. A stable estimate across bandwidths is the reassuring pattern.

Step 5. Sharp versus fuzzy

Here the rule is sharp: crossing 70 determines the scholarship. If crossing 70 only raised the chance of a scholarship (some above 70 decline, some below appeal), the design is fuzzy, and the jump in GPA is divided by the jump in the treatment probability at the cutoff, exactly the Wald ratio from the IV handout with "above 70" as the instrument.

Step 6. Validity checks (slide 114)

No manipulation. Count students per score. If 69 has 20 students and 70 has 45, someone is nudging scores over the line, and the two sides are no longer comparable.

Covariate continuity. Pre-treatment variables (age, prior school GPA, family income) should be smooth through 70. A jump in any of them means something else changes at the cutoff.

Bandwidth and functional form. Show the estimate for bandwidths of 1, 2, 4 and 8 points and for linear and quadratic fits.

Step 7. What the estimate is about

RDD identifies the effect for students near 70. It says nothing about a student who scored 45 or 95. That is the price of the design's credibility: the comparison is almost an experiment, but only at the threshold.

IV versus RDD (slide 115)

	Instrumental variables	Regression discontinuity
Exploits	a nudge that shifts treatment	a threshold that assigns treatment

Estimand	LATE for compliers	local effect at the cutoff
Fragile to	weak instruments, exclusion	manipulation, bandwidth choice
Strong when	a credible instrument exists	the cutoff is real and unforeseeable

Three things to remember

1. The estimate is the jump at the cutoff, from lines fitted on each side, not a comparison of everyone above with everyone below.

2. Check that nobody can game the cutoff and that other variables do not jump there.

3. The answer is local to the threshold.